# SIA101 Reviewer (Weeks 2–7)

SIA101 – System Integration and Architecture 1 · Quezon City University

*Wording follows the slides; obvious typos were fixed. Parts of the stated learning outcomes (stakeholders, functional vs. non-functional requirements, testing tools) are not actually covered in the slides, so they are not included below.*

## WEEK 2 - SIA Requirements

### What is System Integration?

- combining different sub-system components to form a large system
- ensures the functionality of sub-systems

Types: Enterprise Application Integration, Data Integration, Electronic Data Integration

### What is System Architecture?

- a conceptual model that defines a system
- a formal description and representation of a system

*(Slide example: Customer, Buyer's Community, Seller, Clearinghouse, Logistics Service and Payment Service all connected through the Internet.)*

### What is Requirement Elicitation?

- also referred to as "Requirement gathering and specification"
- Guide question: What are the different methodologies in requirement elicitation?

### Methodologies in Requirement Elicitation

1. **Brainstorming** - Groups generate ideas.
2. **Document Analysis** - Study existing documentation.
3. **Focus Group** - Guided group discussion.
4. **Interface Analysis** - Study how system/user interact.
5. **Observation** - Watch users perform tasks.
6. **Interview** - One-on-one questioning.
7. **Prototyping** - Build a simple model.
    - Types of Prototyping: a. Rapid Throwaway Prototype, b. Evolutionary Prototype, c. Incremental Prototype, d. Extreme Prototype
8. **Requirements Workshop** - Structured session with stakeholders.
9. **Reverse Engineering** - Analyze an existing system to derive requirements
10. **Surveying** - Collect data via structure questions.
    - Methods in Surveying:
    - a. Communicating surveys responses and objective
    - b. Be aware of the survey population and its characteristics
    - c. Keep survey short
    - d. Make sure that survey content is clear
    - e. Avoid negative questions
    - f. Avoid complex concepts
    - g. Elicit more details
    - h. Avoid questions that may put respondents on the defensive.

*Note: the slide's technique diagram lists Brainstorming, Document Analysis, Focus Group, Interface Analysis, Interviews, Observation, Process Modeling, Prototype, Requirement Workshops, and Surveys / Questionnaire (Process Modeling appears there, Reverse Engineering appears in the numbered list).*

# WEEK 3 - SIA Requirements

*(Use Case Model, Modeling Tools and Methodologies, Testing Tools and Methodologies)*

## Use Case Model

What is a Use Case Model?

- A model that shows how the different types of users interact with the system to solve a problem.
- Unified Modeling Language (UML)
- A form of system/Software requirements for a new software program

*(Slide example: ATM use cases – Bank Customer: Withdraw Cash, Transfer Funds, Deposit Funds; Maintenance Person: Refill Machine; Bank as an external actor.)*

Purpose of Use Case Diagram a. Specifying the context of a system b. Capturing system requirements c. Validating system architecture d. Implementing and generating test cases e. Developed by analysts and other experts

## Use Case Diagram - Core Concepts

| Element | Meaning |
|---|---|
| Actor | A role that interacts with the system. |
| Use Case | A goal the system fulfills for an actor. |
| System Boundary | The rectangle separating what's inside the system from the actors outside it. |
| <<include>> | A **mandatory** sub-behavior |
| <<extend>> | **Optional/Conditional** behavior inserted at a defined point. |

## SIA Modeling Tools and Methodologies

What is Business Process Modelling?

- it is mainly used to map a work flow
- helps us to visualize work flow or process and make better decisions

Techniques in business process modeling

1. **Business Process Modelling Notation (BPMN)** – the graphical representation of your business process using standard objects. Basic Building Blocks: Flow objects, Connecting objects, Swimlanes, Artifacts.
2. **Unified Modeling Language (UML) Diagrams** – a more modern approach to modelling and documenting processes; offers a flexible and powerful way to visualize almost any business processes.
3. **Flowchart Technique** – most popular diagram; it uses basic shapes and arrow to define relationships, such as processes, decisions or data.

4. **Data Flow Diagram (DFD)** – shows the flow of data or information from one place to another.
5. **Role Activity Diagram (RAD)** – used to map out the intangible roles or ideas of behaviour that are desired within the company.
6. **Role Interaction Diagram (RID)** – illustrates the interaction of various processes with each other within a system. (Examples: Sequence Diagram, Collaboration Diagram)
7. **Gantt Charts** – it relates a list of activities to a time scale; project's timeline.
8. **Integrated Definition for Function Modelling (IDEF)** – a family of methods that support a paradigm capable of addressing the modelling needs of an enterprise and its business areas. (Parts of a function: Input, Control, Output, Mechanism/Resources.)
9. **Coloured Petri Nets (CPN)** – CPN are a graphical oriented language for design, specification, simulation and verification of systems.
10. **Object Oriented Methods** – more than just modelling with objects: it encompasses message-passing, encapsulation, inheritance from class to subclass, and polymorphism.

### Business Analysis Tools

Why do the business analysts need the best business analysis tools for business?

1. To track requirements
2. To manage requirements
3. To describe requirements in detail
4. To model requirements diagrammatically
5. To collaborate with teams and stakeholders

Different types of business analysis tools

1. Requirement-related tools
2. Modelling Tools
3. Collaboration Tools

## WEEK 4 - SIA Requirements

*(Project Life-Cycle Phases, and Other Developmental Phases)*

### Project Management Life Cycle

- developed by **Project Management Institute (PMI)**
- includes Initiation, Planning, Execution, Performance Monitoring/Controlling, and Project Close

| Phase | Key Items |
|---|---|
| 1. Conception & Initiation | Project Charter, Project Initiation |
| 2. Definition & Planning | Scope & Budget, Work Breakdown Schedule, Gantt Chart, Communication Plan, Risk Management |
| 3. Launch or Execution | Status & Tracking, KPIs, Quality, Forecasts |
| 4. Performance & Control | Objectives, Quality Deliverables, Effort & Cost Tracking, Performance |
| 5. Project Close | Post Mortem, Project Punchlist, Reporting |

## PHASE 1: Project Initiation

1. **Business Case** – Explain why the project is necessary and how it will succeed
2. **Feasibility Study** – Research the reason for the project and determine if it will succeed
3. **Project Charter** – How will the project be structured and executed?
4. **Team** – Find the people with the right skills and experience to execute the project
5. **Project Office** – Where the project manager and support staff are located to assist with projects
6. **Review** – Review the initiation phase and keep reviewing progress throughout the project

## PHASE 2: Project Planning

This phase is key to successful project management and focuses on developing a roadmap that everyone will follow.

Two methods used for setting goals: SMART and CLEAR

### SMART Method

- **Specific** – Be clear and specific so your goals are easier to achieve. This also helps you know how and where to get started!
- **Measurable** – Measurable goals can be tracked, allowing you to see your progress. They also tell you when a goal is complete.
- **Actionable** – Are you able to take action to achieve the goal? Actionable goals ensure the steps to get there are within your control.
- **Realistic** – Avoid overwhelm and unnecessary stress and frustration by making the goal realistic.
- **Timebound** – A date helps us stay focused and motivated, inspiring us and providing something to work towards.

### CLEAR Method

- **Collaborative**: Is everyone involved?
- **Limited**: Have you set time and financial boundaries?
- **Emotional**: Does your goal engage all team members?
- **Appreciable**: Is your goal measurable?
- **Refinable**: Can it be adjusted if necessary?

Documents needed to ensure that projects are on track:

- **Scope Statement** – defines the business need, benefits of the project, objectives, deliverables, and key milestones. (Template fields: Project Name, Project Sponsor, Project Manager, Date of Project Approval, Last Revision Date, Scope Description, Project Deliverables, Acceptance Criteria, Constraints, Assumptions.)
- **Work Breakdown Structure (WBS)** – hierarchical and incremental decomposition of the project into phases, deliverables, and work packages. (Slide example: "New Toy for 5-9 year olds" broken into Market Research, Product Design, Product Development, Production Planning, Marketing, Project Management.)
- **Milestones** – a specific point within a project's life cycle used to measure the progress toward the ultimate goal.
- **Gantt Chart** – help you assess how long a project should take, determine the resources needed, and plan the order in which you'll complete tasks.

- **Communication Plan** – a policy-driven approach to providing stakeholders with information. (Table columns: Description, Frequency, Method, Audience, Owner.)
- **Risk Management Plan** – a written document that details the organization's risk management process. (Slide rates risk from 5 Severe down to 1 Minimal across Technical Performance, Cost, and Schedule.)

## PHASE 3: Project Execution

Phase where deliverables are developed and completed.

Tasks completed during the Execution Phase include:

- Develop team
- Assign resources
- Execute project management plans
- Procurement management if needed
- PM directs and manages project execution
- Set up tracking systems
- Task assignments are executed
- Status meetings
- Update project schedule
- Modify project plans as needed

## PHASE 4: Project Performance / Monitoring

Measuring project progression and performance and ensuring that everything happening aligns with the project management plan

Key Performance Indicator (KPI)

1. **Project Objectives** – included measuring of schedule and budget
2. **Quality Deliverables** – determines if deliverables are met
3. **Effort and Cost Tracking** – informs if a project will meet its date of completion
4. **Project Performance** – monitors changes in the project

## PHASE 5: Project Closure

Formal recognition of the completion of a project — everyone agrees that it is completed

Steps to the Project Closure Process

- Transfer all deliverables
- Complete the contracts
- Have a retrospective meeting
- Disband the team
- Document all learnings

# WEEK 5 - SIA Organizational Context

*(Business Processes, IT Environment, Organizational Culture)*

**What is a Business Process?**

- It is a series of steps performed by a group of stakeholders to achieve a concrete goal.
- collection of linked tasks which find their end in the delivery of a service or product to a client.

**Importance of Business Process**

1. To identify important tasks
2. To improve efficiency
3. To streamline communications
4. To set approvals
5. To prevent chaos
6. To standardize procedures

**7 Steps of Business Process Lifecycle**

1. **Define your goals** – know the purpose of the process.
2. **Plan and map your process** – determine the strategies needed in achieving goals.
3. **Set actions and assign stakeholders** – distribute individual tasks to team members and machines.
4. **Test the process** – run the process on a small scale environment and monitor its performance.
5. **Implement the process** – live environment
6. **Monitor the results** – review and analyze processes
7. **Repeat** – replicate the processes

**Business Process Integration (BPI)**

- allows for automation of business processes, integration of systems and services, and secure sharing of data across numerous applications.
- BPI is essential for businesses, looking to connect systems and information efficiently.

## IT Environments

**IT Environment** – it is an integrated collection of technology components that serves the needs of its users and the owner of the resulting system.

**IT Environment Components**

- Computing Platforms
- Applications/Apps
- Connectivity (Networking)
    - Wireless Network (WiFi, 3G/4G/5G data services, Bluetooth)
    - Wired Network (Twisted copper pairs, Fiber optic)
    - Personal Network (Bluetooth)
    - Local Network (Ethernet)
    - Distance (Wide-area - ex. DSL)

**IT Environment Purposes**

- **NEEDS:** Functionality of environment must address the purpose; Cost of environment must be affordable; Reliability of environment must be reasonable
- **RULES:** Laws and regulations influence how an owner operates their IT environment; Owners or their representatives may impose their own rules to ensure purpose is achieved.
- **CHOICES:** Owner's Needs; Rules that must be met; Available affordable Technology

## Organizational Structure

**What is an Organizational Structure?**

- A system that outlines how certain activities are directed in order to achieve the goals of an organization.
- These activities can include the following: rules, roles, responsibilities
- It also determines how information flows between levels within an organization.

**Understanding Organizational Structure**

- In business, organization structure is defined as a specific hierarchy within the organization.
- It defines each employee's job to fit in the system
- It is normally illustrated in some sort of chart or diagram like a pyramid.

**Centralized VS Decentralized Organizational Structure**

- **Centralized** – hierarchical decision making; different branches have similar processes; common in large organization
- **Decentralized** – democratic decision making; different branches adapts their specific needs; common in small organization

**Types of Organizational Structure**

1. **Functional Structure -** Bureaucratic organizational structure; dividing the firm into different departments to have uniformity; centralized organizational structure.
    a. Structure: Board of Directors → Finance, Marketing, HR, Admin, Etc.
    b. Advantages:
        i. Experience, Specialist Staff, Standardisation, Greater Communication, Economies of Scale, Efficiency.
    c. Disadvantages:
        i. Lack of Variety and & Job Enrichment, Limits Flair & Ingenuity, Greater Bureaucracy & Form Filling, Compromise on Functional Requirements.
2. **Divisional or Multidivisional Structure -** Second most common in large companies; a parent organization owns subsidiaries that use the same name and brand.
    a. Structure**:** Board of Directors → Chemicals/Plastic/Electronic Division (or geographic: European/US/Rest of World)
    b. Advantages:
        i. Senior Management Time Freed Up, Greater Local Knowledge, Motivation through greater decision powers, Greater Understanding, Speed

    c. Disadvantages:
        i. Duplication of Service/Cost, Lack of Specialisation, Extra Processing, Coordination more Difficult
3. **Flatarchy Structure** - The third type and is used among many startups; it flattens the hierarchy and chain of command and gives its employees a lot of autonomy.
4. **Matrix Structure** - There is more than one line of reporting managers; solid line or dotted line reporting.

# Week 6 - SIA Acquisition and Sourcing

*(Build and Buy, In-Sourcing)*

## Acquisition

### Procurement Sourcing and Acquisition Process

- **Request for procurement** is documented in a **purchase requisition order** and submitted in the procurement unit to be managed in accordance with the following guidelines.

### Sourcing

- it includes activity to identify and assess potential suppliers, selecting and engaging with best suppliers.

### ACQUISITION: HARDWARE and SOFTWARE

- requirements analysis
- request for proposal
- vendor selection

### Sourcing Requirements

- **Market Survey:** Quantities Available; Quality; Pricing; Source and origin of the items; Technical Specifications
- **Determining Source of Supply:** assess sources (local, national, and international). The first principle should be to source all commodities locally and in-country, as close to the field as possible to reduce transport time and cost.
- **Sourcing Supply Internationally:** identify international vendors and/or manufacturers for the materials unavailable locally or nationally in the quantities required.
    - details of pre-positioned items available at a CARE International level
    - information and contact details of known international suppliers of humanitarian relief items
    - details of vendors who have signed pre-supply agreements with the CO or CARE Members
    - check with other NGOs to find out who they are using as their emergency suppliers.

### Vendor Selection and Screening

- **Methods for vendor selection** include: sole or single sourcing; request for quotations; sealed bidding process
- **Sole Source** – only one source is available (i.e. local water companies)

- **Single Source** – organizations select one supplier; there are more than one supplier available

**Request for Quotations or Sealed Bids**

- The requirement for quotes or sealed bids will be determined by the bidding thresholds outlined in **CAREs CI Harmonization Procurement Policy**.
- The procurement unit should prepare and issue a **request for quotation (RFQ)** for the goods and services required.
- RFQs and vendor responses should be tracked using Tracking Sheet for Sealed Bids.
- The procurement unit should coordinate the analysis of bids and selection of vendors.
- Processes for the review of bids (for example bid committees) should be streamlined for the period of the emergency. Follow CAREs CI Harmonization Procurement Policy.

**Vendor Screening**

- When selected as the preferred supplier, vendors must be screened before CARE confirms an order or contract with them.
    - request potential vendors and suppliers to complete a vendor questionnaire
    - check references, preferably from other NGOs
    - screen all potential vendors against the anti-terrorism and debarred listings, for both individuals and organizations.

### In-Sourcing

**What is Insourcing?** – it is the assignment of a project to a person or a department within the company rather than a third-party.

**Advantages**

- Competitive Advantage
- Trade Secrets
- Organizational Structure
- Direction & Control
- Cost

**Disadvantages**

- Cost
- Capabilities
- Focus

## WEEK 7 - SIA Acquisition and Sourcing

*(Outsourcing)*

### Outsourcing

**What is Outsourcing?** – It is a business practice in which services or job functions are farmed out to a third party. *Types: Onshore, Nearshore, Offshore*

**Benefits of Outsourcing**

- lower costs (due to economies of scale or lower labor rates)
- increased efficiency
- variable capacity
- increased focus on strategy/core competencies
- access to skills or resources
- increased flexibility to meet changing business and commercial conditions
- accelerated time to market
- lower ongoing investment in internal infrastructure
- access to innovation, intellectual property, and thought leadership
- possible cash influx resulting from transfer of assets to the new provider

**Risks of Outsourcing**

- slower turnaround time
- lack of business or domain knowledge
- language and cultural barriers
- time zone differences
- lack of control

**Outsourcing IT Functions**

- **Infrastructure Outsourcing:** service desk; data center outsource; network services; security operations; infrastructure management
- **Application Outsourcing:** new app development; legacy system maintenance; testing and QA services; packaged software implementation and management

**Challenges of Outsourcing**

- Difficult to implement and failure rate remains high: failure rate is between 40-70%; conflict of interest
- Absence of a good business case: "Quick-fix"

**Service Level Agreements (SLA)**

- A contract between an IT services provider and a customer that specifies, usually in measurable terms, what services the vendor will furnish.

**Choosing the Right Outsourcing Provider Portfolio**

- Vendors (outsource) should have cooperation and do not blame each other
- CIO's needs to find qualified staff with the right skills

**How to Select a Service Provider?**

- articulate what you want from the outsourcing relationship
- analyze important criteria to know what is the best for your organization.
- Questions to Consider:
    - What's more important to you: the total amount of savings an outsourcer can provide you or how quickly they can cut your costs?
    - Do you want broad capabilities or expertise in a specific area?
    - Do you want low, fixed costs or more variable price options?